

AI - Based Intrusion Detection System

Gokul Athikavil Shaji¹, Saathvika Prasad²

¹ gokul.shaji@studio.unibo.it

² saathvika.prasad@studio.unibo.it

1 Introduction

With the rapid evolution of cyber threats, traditional security mechanisms have become increasingly inadequate. Cyber attackers continuously develop sophisticated techniques to evade detection, necessitating more advanced security solutions. Intrusion Detection Systems (IDS) are crucial in identifying and mitigating these threats by monitoring network traffic and detecting anomalies. However, conventional IDS often suffer from high false positive rates, difficulty in distinguishing benign anomalies from actual threats, and limited adaptability to zero-day attacks. This necessitates the development of an improved IDS framework that can efficiently detect and classify cyber threats while minimizing errors.

This project introduces a multi-stage IDS framework that leverages machine learning techniques for enhanced detection and classification of cyber threats. The system is structured into three stages: anomaly detection, attack classification, and zero-day attack detection. Each stage is optimized using hyperparameter tuning and threshold selection to maximize accuracy and minimize false positives.

2 Preprocessing

2.1 Quantile Transformation

Quantile transformation is a preprocessing technique used to transform the distribution of data to follow a desired distribution, such as a normal or uniform distribution. This process is essential for ensuring that the data is well-prepared for machine learning models, which often assume that the input data follows a specific distribution.

- **Objective:** Improve Model Performance: The primary goal of quantile transformation is to improve the performance of machine learning models. By transforming the data distributions, the models can better learn patterns and make more accurate predictions. This is particularly important when dealing with skewed or non-normal data, which can negatively impact model performance.
- **Method:** Apply the `QuantileTransformer`, from the `sklearn.preprocessing` module is used to perform the transformation. This transformer maps the data to a uniform distribution by ranking the data and scaling it to the desired output distribution.
- **Output Distribution:** Depending on the dataset's characteristics and the requirements of the machine learning model, the `QuantileTransformer` can be configured to transform the data to either a normal distribution or a uniform distribution. This flexibility allows for better alignment with the model's assumptions and improves its overall performance.

By implementing quantile transformation, the data becomes more suitable for analysis and modeling, leading to more robust and reliable machine learning outcomes.

2.2 Aligning Features with Datasets

Aligning features with datasets is essential to ensure that the preprocessing steps are consistent and relevant to the datasets used in your project. This step involves verifying that the features being used for preprocessing match those present in the datasets, either exactly or closely enough to maintain data integrity and relevance.

- **Objective:** Maintain Consistency Across Datasets: Ensuring that the same feature set is used across all datasets helps maintain uniformity and avoids discrepancies that could affect the analysis and modeling. This consistency is crucial for the reproducibility and reliability of your results.
- **Method:** Cross-check Features: Carefully compare the features in the preprocessing step with those in the datasets. This involves:
 - **Feature Matching:** Ensure that each feature in the preprocessing set has a corresponding feature in the datasets.
 - **Relevance Check:** Verify that the features are relevant to the analysis and contribute meaningfully to the model.
 - **Data Integrity:** Maintain the integrity of the data by avoiding the inclusion of features that are not present in all datasets, as this could lead to inconsistencies and errors.

By aligning the features with the datasets, you ensure that your preprocessing steps are applicable and beneficial to the data, leading to more accurate and reliable results.

2.3 Removing Irrelevant or Redundant Features

Identifying and removing irrelevant or redundant features is a crucial step in data preprocessing. This step simplifies the data, making it easier to analyze and process, and improves the efficiency and performance of machine learning models.

- **Objective:** Simplify the Data and Improve Processing Efficiency: By removing features that do not contribute to the analysis or are redundant, you can reduce the complexity of the dataset. This simplification leads to faster processing times and more efficient use of computational resources, and it helps prevent overfitting by reducing the risk of the model learning noise instead of meaningful patterns.

3 Methodology

3.1 Stage 1: Anomaly Detection (Unsupervised Learning)

3.1.1 Objective

In the first stage of the anomaly detection system, the primary goal is to accurately distinguish between benign (normal) and anomalous (potentially malicious) network traffic. This stage leverages unsupervised learning techniques to analyze network data without relying on labeled samples. The key challenge is to achieve high recall, ensuring that most anomalies are detected, while maintaining an acceptable false positive rate, which avoids incorrectly flagging benign traffic as anomalous. By carefully balancing high recall and an acceptable false positive rate, the first stage of anomaly detection creates a robust foundation for the subsequent stages of the intrusion detection system. This ensures that the system effectively identifies potential threats while minimizing the impact of false positives.

3.1.2 Training Process

1. Data Preparation:

- **Exclusive Training on Benign Data:** The model is trained exclusively on benign data to learn the patterns of normal behavior. By focusing on benign data, the model establishes a baseline of what constitutes normal traffic, enabling it to identify anomalies effectively.
- **Anomaly Detection:** During prediction, any significant deviation from the benign data distribution is flagged as anomalous. This approach ensures that the model can detect previously unseen or zero-day attacks.
- **Datasets:** The datasets used for training and evaluation are CICIDS2017 and CICIDS2018. These datasets provide a comprehensive collection of benign and malicious network traffic, enabling robust anomaly detection model training.

2. Hyperparameter Tuning:

- **Optimization Framework:** Before selecting the best model, we optimized each model's hyperparameters using **Optuna**, a hyperparameter optimization framework.
- **Objective Function:** The objective function was designed to maximize the Area Under the Receiver Operating Characteristic Curve (AUROC).
- **Why AUROC?** Unlike F-scores, which require a fixed threshold for calculation, AUROC evaluates the model's ability to distinguish between classes across all possible thresholds. This provides a comprehensive understanding of the trade-off between precision and recall without biasing the model selection toward any specific threshold.
- **Ensuring Optimal Performance:** The tuned hyperparameters ensure that each model performs optimally before being compared for Stage 1 suitability.

3. Model Mechanism:

- **Anomaly Scoring:** The trained model assigns an anomaly score to each input sample, where higher scores indicate greater deviation from normal patterns. This scoring mechanism allows the model to quantify the degree of anomaly for each sample.

4. Decision Mechanism:

- **Score Calculation:** For OC-SVM, the `model.decision_function` is used to calculate scores. These scores are negated to align the system with the desired behavior:
 - **Significant Deviations:** Higher anomaly scores represent more significant deviations (potential attacks).
- **Threshold Determination:** A benign threshold is determined post-training. If the anomaly score of a sample is:
 - **Greater than the Threshold:** The sample is flagged as malicious and passed to the next stage.
 - **Less than or Equal to the Threshold:** The sample is classified as benign.

3.1.3 Why OC-SVM is Selected

OC-SVM was selected for the following reasons:

1. Balanced Performance:

- OC-SVM has the highest F1 score (0.4175) among all models, indicating a good trade-off between precision and recall. The F1 score is a metric that combines both precision and recall, providing a single measure of a model's performance. By achieving the highest F1 score, OC-SVM demonstrates its effectiveness in accurately identifying both benign and anomalous samples.

2. High Recall:

- Recall is critical to ensure that most attacks are passed to the next stage. OC-SVM achieves a recall of 37.39%. Recall, also known as sensitivity or true positive rate, measures the proportion of actual anomalies that are correctly identified by the model. High recall is essential to minimize the risk of undetected threats, ensuring that the system captures the majority of malicious activities.

3. Moderate Precision and FPR:

- OC-SVM's precision is 47.25%, and its false positive rate (FPR) is 2.19%. Precision measures the proportion of correctly identified anomalies among all flagged samples, while FPR measures the proportion of benign samples that are incorrectly flagged as anomalous. By balancing moderate precision with a low FPR, OC-SVM ensures that the system is neither overwhelmed with false positives nor too lenient in attack detection.

4. Hyperparameter Optimization:

- The hyperparameters for OC-SVM were carefully tuned using Optuna, a hyperparameter optimization framework, ensuring optimal AUROC performance. The Area Under the Receiver Operating Characteristic Curve (AUROC) evaluates the model's ability to distinguish between classes across all possible thresholds, providing a comprehensive understanding of the trade-off between precision and recall. The optimized hyperparameters enhance the model's robustness in distinguishing between benign and malicious data, leading to more reliable and accurate predictions.

Model	F1 Score	Precision	Recall	False Positive Rate (FPR)
Autoencoder	0.385	0.6092	0.2814	0.0095
OC-SVM	0.4175	0.4725	0.3739	0.0219
Elliptic Envelope	0.2184	0.1365	0.5452	0.1816
Isolation Forest	0.3264	0.4786	0.2477	0.0142

Table 1: **Model Evaluation**

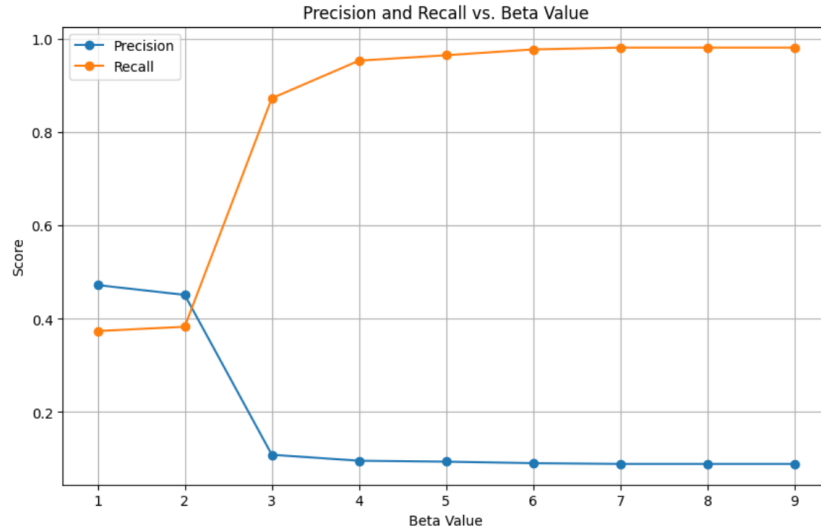


Figure 1: Precision and Recall vs Beta Value

3.1.4 Threshold Selection: Benign Threshold

Once OC-SVM was selected, the next step was to determine the benign threshold. The threshold is crucial as it decides which samples are flagged as malicious and passed to the next stage.

1. Threshold Evaluation:

- **Metrics Used for Evaluation:** Thresholds were evaluated using key metrics such as precision, recall, and false positive rate (FPR). The objective was to select a threshold that maximizes recall without excessively increasing the FPR.
 - **Precision:** Precision measures the proportion of correctly identified malicious samples among all flagged samples. A high precision indicates that the system is accurate in identifying true positives, minimizing false alarms.
 - **Recall:** Recall, also known as sensitivity, measures the proportion of actual malicious samples that are correctly identified. High recall is crucial to ensure that most attacks are detected and not missed by the system.
 - **False Positive Rate (FPR):** FPR measures the proportion of benign samples that are incorrectly flagged as malicious. An acceptable FPR ensures that the system does not overwhelm subsequent stages with false positives, maintaining efficiency.

2. F4 Threshold Selection:

- **Chosen Threshold:** The F4 threshold (-0.003660) was chosen based on its performance across the evaluation metrics.
 - **High Recall:** The F4 threshold provides a high recall of 95.41
 - **Moderate Precision:** The threshold maintains a moderate precision of 9.56
 - **Manageable False Positive Rate:** The F4 threshold keeps the FPR at 47.54

Metric	Value	Threshold	Precision	Recall	FPR
F1	0.4175	0.000476	0.4725	0.3739	0.021987
F2	0.3951	0.000353	0.4515	0.3831	0.024520
F4 (Selected)	0.6243	-0.003660	0.0956	0.9541	0.475368
F5	0.711	-0.003840	0.0936	0.9657	0.492289

Table 2: Metrics for F4 Threshold

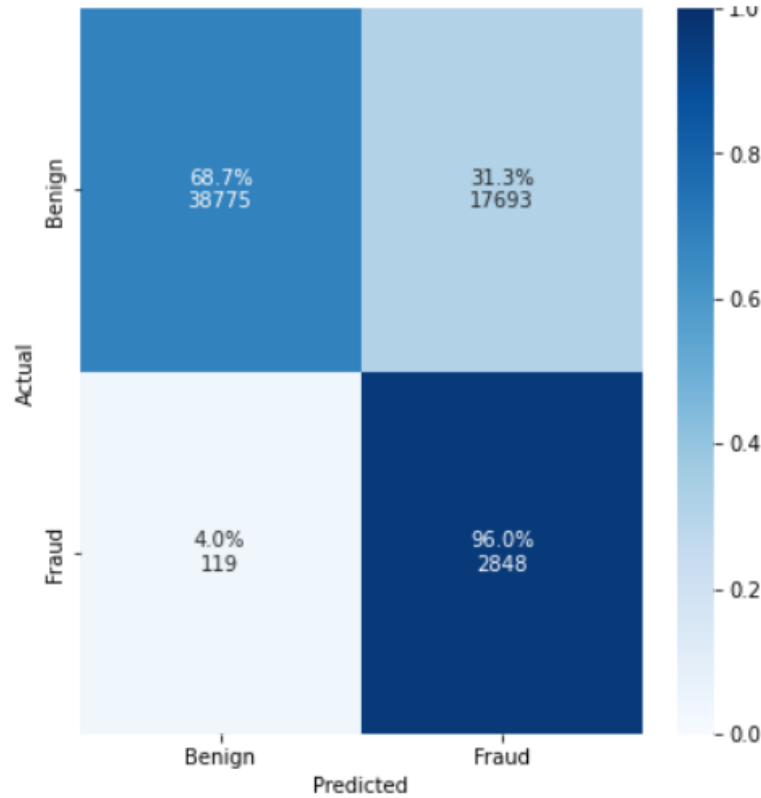


Figure 2: Confusion Matrix - Stage 1

3.2 Stage 2: Attack Classification (Supervised Learning)

3.2.1 Overview

Stage 2 of the Intrusion Detection System (IDS) is designed to classify anomalies detected in Stage 1 into specific attack types. This stage evaluates several supervised machine learning models to identify the best attack classifier based on performance metrics. Hyperparameter tuning was conducted using Optuna, optimizing for f1_weighted to ensure robust handling of imbalanced attack data. The evaluation focuses on key metrics, including F1 macro, F1 weighted, accuracy, balanced accuracy, precision, and recall.

- **Supervised Machine Learning Models:** Several supervised learning models are evaluated to determine the most effective classifier for attack detection. These models include:
 - **Feedforward Neural Network (FNN):** A neural network model that learns complex patterns in the data through multiple layers of interconnected nodes.
 - **Support Vector Machine (SVM):** A model that finds the optimal hyperplane to separate different classes in the data.

- **Random Forest:** An ensemble model that combines multiple decision trees to improve classification performance.
- **XGBoost:** An advanced gradient boosting model that focuses on improving prediction accuracy through iterative training.
- **Hyperparameter Tuning:** Hyperparameter tuning is a critical process to optimize the performance of machine learning models. Using **Optuna**, hyperparameters for each model are carefully tuned to maximize the F1 weighted score, ensuring robust handling of imbalanced attack data. This involves:
 - **Objective Function:** The objective function is designed to maximize the F1 weighted score, which combines both precision and recall while accounting for class imbalance.
 - **Optimization Process:** Optuna’s optimization process explores different hyperparameter configurations, evaluating the model’s performance on the training data to identify the best hyperparameters.

Model	F1 Macro	F1 Weighted	Accuracy (%)	Balanced Accuracy (%)
FNN	0.6428	0.9139	87.63	88.31
SVM	0.7527	0.9536	93.76	94.33
Random Forest	0.9094	0.9870	98.46	96.54
XGBoost	0.8921	0.9756	96.78	95.31

Table 3: Model Evaluation

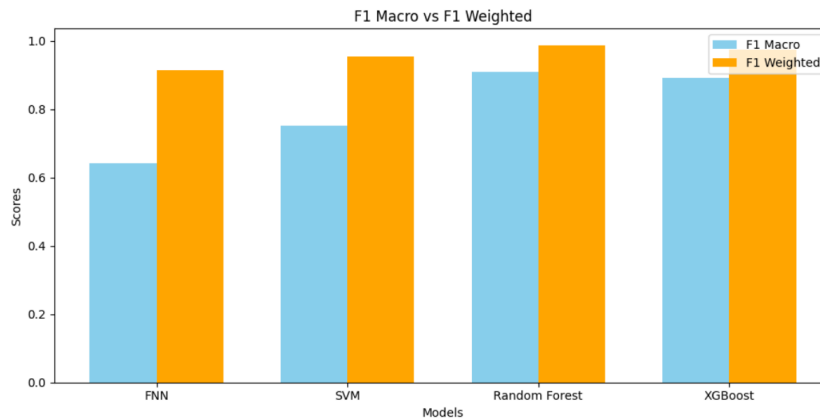


Figure 3: F1 Macro vs F1 Weighted

3.2.2 Threshold Optimization: Confidence Threshold

The threshold is optimized using a custom Threshold Evaluation Loop. This process is designed to find the optimal confidence threshold that maximizes the model’s performance in classifying attack types.

1. Predict Classes:

- For each threshold value, predicted probabilities of the samples are compared.
- If the highest predicted probability for a sample exceeds the threshold, the sample is classified as the corresponding attack type.

- Otherwise, the sample is classified as 'Unknown.'

2. Calculate F1 Scores:

- The macro and weighted F1 scores are calculated for each threshold value.
- **Macro F1 Score:** This is the unweighted average of the F1 scores for each class, treating all classes equally.
- **Weighted F1 Score:** This is the average of the F1 scores for each class, weighted by the number of true instances for each class. This score accounts for class imbalance.

3. Update Best Scores:

- If the current threshold value yields a higher F1 score than the previously recorded best score, the best score and threshold are updated.
- This iterative process continues until the optimal threshold value is identified.

Threshold for Random Forest:

- The optimal confidence threshold for the Random Forest model was determined to be **0.96**.
- **Significance:** This threshold ensures a balance between high precision and recall, optimizing the model's ability to correctly classify attack types while minimizing false positives.

Threshold for Random Forest:

- The optimal confidence threshold for Random Forest is 0.96.

3.2.3 Feature Enhancement

To improve the accuracy of the attack classification model in Stage 2, anomaly scores from Stage 1 were incorporated as additional features. This approach helps to reduce noise in the data and enhances the robustness of the model. By leveraging the anomaly scores, the model can better differentiate between benign and malicious activities, leading to more accurate predictions.

- **Incorporation of Anomaly Scores:** The anomaly scores generated in Stage 1 provide valuable information about the likelihood of a sample being anomalous. By including these scores as features in Stage 2, the model gains additional context, improving its ability to classify attacks accurately.
- **Noise Reduction:** Adding anomaly scores helps to reduce noise in the data by providing a clearer distinction between normal and anomalous samples. This leads to more robust and reliable model performance.
- **Improved Robustness:** The incorporation of anomaly scores enhances the model's robustness, making it more resilient to variations in the data. This results in better generalization and reduces the risk of overfitting.

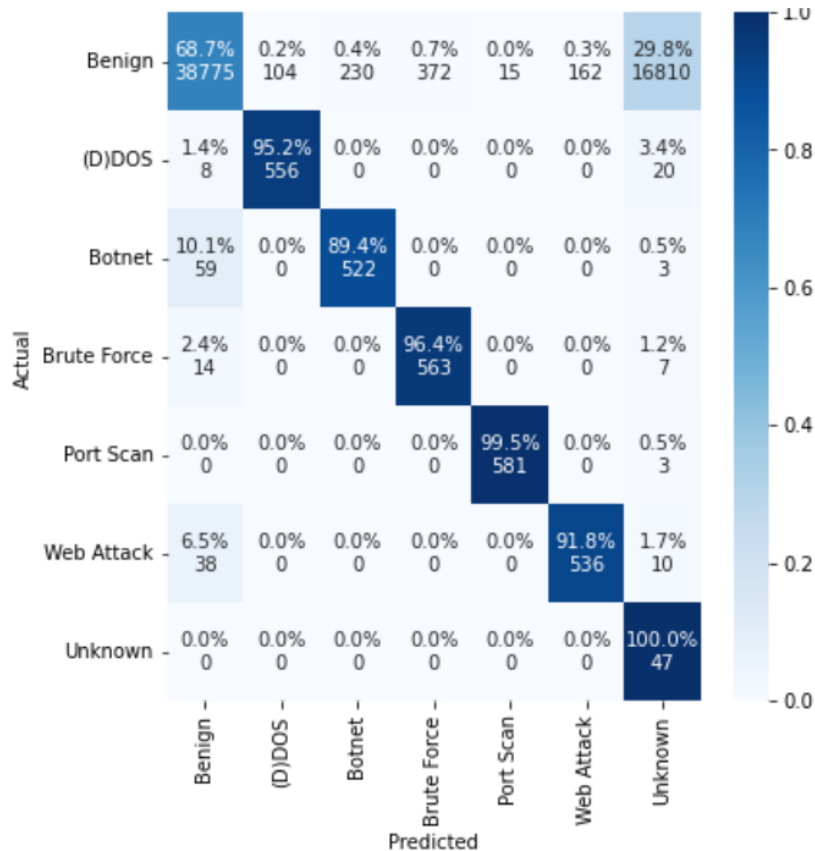


Figure 4: Confusion Matrix - Stage 2

3.2.4 Final Model Selection: Random Forest

Random Forest was selected as the final model for Stage 2 due to its superior performance across all evaluation metrics. This model demonstrated exceptional precision and recall, which are critical for minimizing false positives and detecting a wide range of attacks.

- **Superior Performance:** Random Forest outperformed other models in terms of precision and recall, making it the best choice for attack classification. High precision ensures that the model accurately identifies true positives, while high recall ensures that most attacks are detected.
- **Incorporation of Anomaly Scores:** The addition of anomaly scores as features further enhances the robustness, generalization, and resistance to overfitting of the Random Forest model. This integration provides a more comprehensive understanding of the data, leading to better classification performance.
- **Confidence Threshold:** A confidence threshold of 0.96 was established to ensure reliable classification. If the highest predicted probability for a sample exceeds this threshold, the sample is classified as the corresponding attack type. Otherwise, the sample is classified as 'Unknown' for further analysis. This approach helps to handle uncertain samples effectively.
- **Minimizing False Positives:** The high precision and recall of the Random Forest model are crucial for minimizing false positives, ensuring that the system is not overwhelmed with incorrect alerts. This balance is essential for maintaining the efficiency and reliability of the intrusion detection system.

3.3 Stage 3: Zero-Day Attack Classification

3.3.1 Objective

The goal of Stage 3 is to enhance the classification of samples that are potentially zero-day attacks. This stage leverages a threshold-based approach to distinguish between benign and unknown attack samples. By refining the classification process, the system aims to improve its detection capabilities for previously unseen or zero-day

- **Improving Classification:** The primary objective is to accurately classify samples that might be zero-day attacks. Zero-day attacks are previously unknown threats that exploit vulnerabilities not yet identified or patched. By effectively distinguishing these attacks from benign samples, the system can provide an additional layer of security against emerging threats.
- **Leveraging Thresholds:** The classification process involves using thresholds to differentiate between benign and unknown attack samples. The thresholds are determined based on the distribution of data points and their respective quantiles. This approach ensures that the classification is both accurate and reliable.

3.3.2 Threshold Selection

The quantiles tested in Stage 3 are derived from the data points selected in Stage 1. The benign threshold established in Stage 1 is used as a baseline to classify samples as either "Benign" or "Unknown." In Stage 3, these quantiles help determine a range of benign samples between the Benign Threshold and the Zeroday Threshold. Samples that fall within this range are considered benign.

Detailed Explanation:

- **Quantile Selection:** The quantiles tested in Stage 3 are based on the distribution of data points identified in Stage 1. These quantiles provide a statistical basis for determining the thresholds that will be used in the classification process.
- **Benign Threshold:** The benign threshold from Stage 1 is used to classify samples into two categories: "Benign" and "Unknown." This threshold serves as the initial filter to identify potential anomalies.
- **Range Determination:** In Stage 3, the selected quantiles are used to define a range of benign samples. This range lies between the Benign Threshold and the Zeroday Threshold. Samples falling within this range are classified as benign, ensuring that only true anomalies are flagged for further analysis.
- **Optimal Quantile Selection:** After evaluating these quantiles, the 0.99 quantile was found to result in the best accuracy and F1 score. This quantile provides a good balance between identifying zero-day attacks and avoiding false positives. It ensures that the classification process is both precise and comprehensive, enhancing the overall effectiveness of the intrusion detection system.

Quantile	Threshold Value
0.995	0.007645
0.99	0.002157
0.975	0.000296
0.95	-0.000221

Table 4: Quantile and Threshold Value

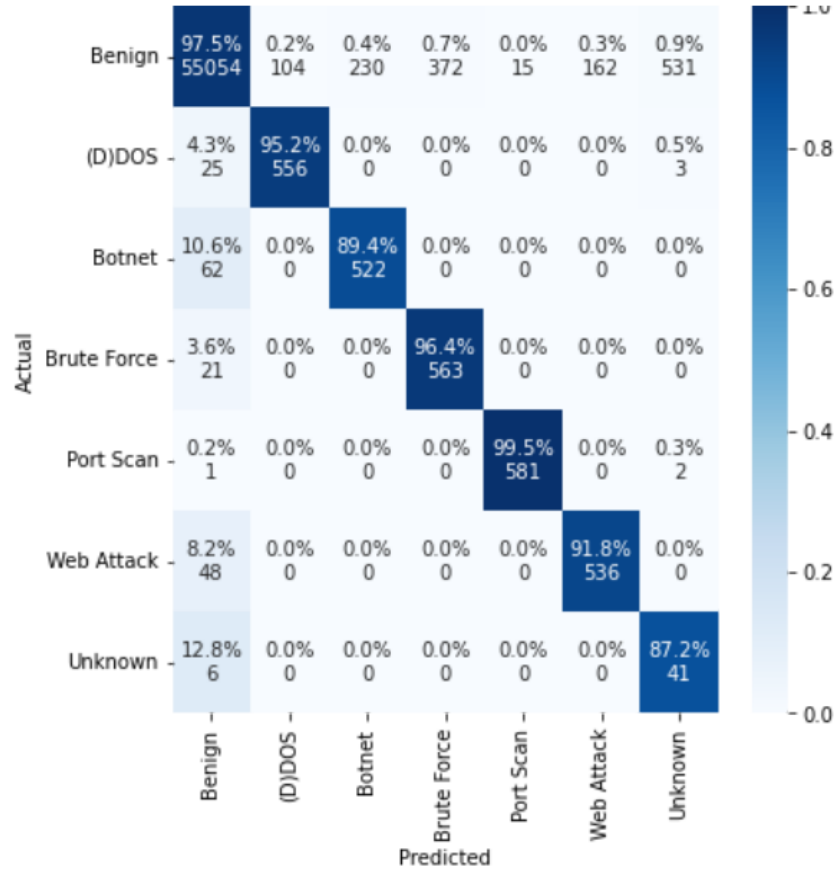


Figure 5: Confusion Matrix - Stage 3

3.3.3 Steps in Stage 3

1. Classifying "Unknown" Samples:

- In Stage 2, the model classifies known attacks and labels the rest as "Unknown" if it is unsure about them.
- Stage 3 focuses on refining the classification of these "Unknown" samples to determine their true nature.
- This refinement is crucial to accurately identify zero-day attacks that might have been misclassified or missed in the previous stage.

2. Applying the Zeroday Threshold:

- The Zeroday Threshold is applied to the predicted probabilities of the "Unknown" samples.
- If the highest predicted probability for a sample exceeds the Zeroday Threshold, the sample is classified as a "Zero-Day Attack."

- If the predicted probability is below the Zeroday Threshold, the sample is classified as "Benign."
- This threshold-based approach ensures that only the most likely zero-day attacks are flagged, reducing false positives and improving detection accuracy.

3. Final Classification:

- Once the Zeroday Threshold is applied, the predictions for "Unknown" samples are updated.
- This step helps to reduce the number of ambiguous predictions, providing clearer classification results.
- The refined classification improves the overall accuracy of the intrusion detection system by ensuring that most zero-day attacks are correctly identified and benign samples are not misclassified.
- By focusing on refining the classification of "Unknown" samples, Stage 3 enhances the system's ability to detect emerging threats and provides a robust defense mechanism against zero-day attacks.

4 Results

In Stage 1, the OC-SVM model efficiently detects anomalies with a recall of 37.39%. This result highlights the model's capability to identify a significant portion of malicious activities while maintaining an acceptable level of false positives. By learning the patterns of normal behavior from benign data, OC-SVM effectively distinguishes anomalous network traffic, contributing to the overall robustness of the intrusion detection system.

In Stage 2, the Random Forest model demonstrates its prowess in classifying attack types, achieving an impressive accuracy of 98.46%. This high level of accuracy indicates the model's robustness in distinguishing between various types of network attacks. Hyperparameter tuning using Optuna further optimizes the model's performance, ensuring it can handle imbalanced attack data. The incorporation of anomaly scores from Stage 1 as additional features enhances the model's ability to classify attacks accurately, improving overall system effectiveness.

Finally, in Stage 3, the application of the 0.99 quantile-based Zeroday Threshold proves to be highly effective in distinguishing zero-day attacks. This approach ensures that previously unseen threats are accurately identified while minimizing the occurrence of false positives. By focusing on refining the classification of "Unknown" samples from Stage 2, the system reduces the number of ambiguous predictions, providing clearer and more reliable classification results. Overall, the combination of these stages provides a comprehensive and robust defense mechanism against evolving cyber threats.

5 Conclusion

This report presents a robust multi-stage IDS framework capable of detecting and classifying cyber threats with high accuracy. The combination of OC-SVM for anomaly detection, Random Forest for attack classification, and a threshold-based approach for zero-day detection ensures a comprehensive defense mechanism against evolving cyber threats. Future work could involve real-time deployment and further model refinements to enhance adaptability against emerging attack vectors.